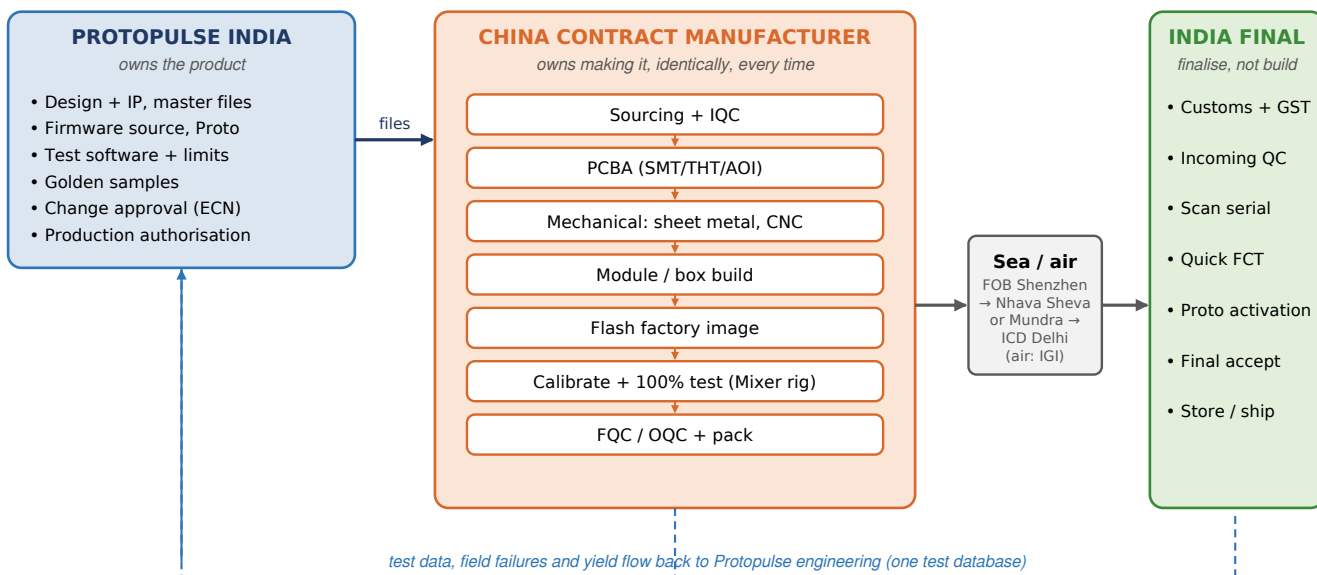


China assembly line master plan

How ProtoPulse gets a qualified Chinese contract manufacturer to build Baker and Mixer identically, ship them to Delhi, and leave India with only final acceptance, light finishing and shipping.



The factory runs Protopulse's process with Protopulse's test rig; Protopulse keeps design, test logic, software activation and final approval.

₹1,44,842

Baker production unit today
(China parts, Delhi build)

₹13,740

Mixer production unit today

≤ \$1,305

Baker turnkey ex-works
price needed to beat today

≤ \$127

Mixer turnkey ex-works
price needed to beat today

The decision this plan recommends. Put **Mixer** into full China turnkey (PCBA, box build, 100% test, packing) from the first production batch. Take **Baker** to China in two steps: first as **tested modules (SKD)** with final integration and PCB-making acceptance in Delhi, then to **full turnkey** once factory quotes show a landed cost below today's ₹1,44,842. The rest of this document is the plan to get there in about 12 months.

Built from: Protopus Cost Report (29 Sep 2026), Protopus Baker Workflow Story, Protopus BOM Baker + Mixer workbook, and the China assembly-line research conversation of 29 Sep 2026. Supplier capabilities are as the companies publicly describe them and must be verified by RFQ, audit and pilot build. Duty, freight and compliance points must be confirmed with a customs broker and a certification consultant.

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1 Executive summary

ProtoPulse has two validated prototypes and a costed production design for each. The goal is a manufacturing system in which a Chinese contract manufacturer reproduces them identically, hundreds or thousands of times, and India becomes a finalisation and distribution centre rather than a second factory.

	Baker (PCB-making machine)	Mixer (8-in-1 instrument)
Prototype cost (India, 1 unit)	₹3,73,418	₹74,847
Production unit today (China parts, Delhi build, batch 100)	₹1,44,842	₹13,740
Deck hardware target	₹1,50,000	₹15,000
Headroom today	₹5,158 (3.4%)	₹1,260 (8.4%)
Product type for a factory	Mechatronic machine: frame, 3 ball-screw axes + 2 Z, spindle, Class 4 laser module, UV/LCD, fluidics, control box	Electronics instrument: one 4-layer PCBA + FPGA module + extruded case
Right kind of factory	Mechatronics contract manufacturer / box-build integrator	EMS / turnkey PCBA + box build
Recommended China scope, year 1	Tested modules (SKD) → full turnkey when quotes justify	Full turnkey from the first batch
What stays in India	Final integration (SKD phase), calibration check, make-a-PCB acceptance, activation, shipping	Incoming check, activation, shipping

The ten things that matter most

- Freeze the design before you RFQ.** Close the four open Baker points (mask cleanliness, film registration, jig shrink, laser drift), pick one board size (100 × 160 vs 100 × 150 mm), and freeze the production electronics (Manta M8P + CB1, or a custom controller).
- Send a manufacturing package, not a prototype.** A factory must be able to build from your files without asking what a part does (§6).
- Use two different kinds of factory.** Mixer is an EMS job; Baker is a mechatronics box-build job. One supplier rarely does both best (§8).
- Price the landed cost, not the factory quote.** For 100 Bakers the factory must hit about \$1,305 ex-works just to match today's Delhi-built cost; parts alone are about \$1,302 at report prices (§12).
- Own the test system.** Mixer + Proto test software becomes the production tester. The factory operates it; Protopulse owns limits, software and data (§10).
- Qualify, then pilot, then commit.** 10–12 RFQs → 3 finalists → 2 EVT suppliers → 1 primary + 1 backup (§8, §9).
- Run EVT → DVT → PVT with numbers.** Every gate has entry and exit criteria and a signed production authorisation (§9).
- Control change.** No part, supplier, process or firmware change without a signed ECN; tooling and fixtures are Protopulse property (§13).
- Protect IP with architecture, not just contracts.** The factory flashes a locked factory image; Proto activation and calibration logic stay with Protopulse (§14).
- Classify for import and compliance early.** HS codes, laser safety for Baker's 10 W 445 nm module, Wi-Fi on the CB1, BIS/WPC/EPR checks, IEC and a customs broker, before PVT (§15, §16).

2 The products being industrialised

Baker: what the factory is actually building

Baker turns blank copper-clad FR4 into an assembled board in 15 steps on one shuttle (see the Workflow Story). For a factory it is a machine with five manufacturing modules:

Module	Content (production BOM)	Build skill needed	Critical characteristics
A Chassis	Sheet-metal frame, acrylic panels, orange 450 nm laser-safe window, door interlocks, E-stop	Sheet metal, laser cutting, bending, powder coat	Squareness, light-tightness around laser zone, interlock function
B Motion	3 × SFU1605 ball screw + BK/BF12, 3 × HGR15 rail pair, 3 × NEMA23, 8 × NEMA17, 2 × T8 Z axes	Precision mechanical assembly	Rail parallelism, squareness, backlash, homing repeatability
C Process heads	500 W ER11 spindle + touch probe; 10 W 445 nm laser + air assist; 10.1" 12K mono LCD + ChiTu board + 405 nm UV board; syringes, peristaltic pump, air knife; Kapton film, 4 electromagnets, solenoid; vacuum pick, hotplate	Electromechanical sub-assembly, optics alignment	Spindle runout, laser-to-spindle offset, UV image registration, film registration ±50 µm
D Electronics	BTT Manta M8P V2 + CB1, TMC2209 drivers, HDMI7 screen, Mean Well LRS-350-24, SSR, thermocouple, harness	Control-box wiring, harness making, flashing	Earth bond, insulation, correct firmware image, interlock chain
E Protopulse-specific	Firmware, Proto UI, calibration data, test software, serial number, activation	Operated by factory; owned by Protopulse	Locked factory image; data upload to Protopulse database

Mixer: what the factory is actually building

Block	Production content	Build skill needed
Main PCBA	4-layer board: Tang Nano 9K FPGA module, AD9288 ADC, 2 × DAC908, FT232HL USB bridge, ADS1256 DMM, 4 × AD8066, REF5025, supply DACs, INA226, relays, protection	SMT (fine-pitch), AOI, THT for connectors
I/O	BNC, banana jacks, USB-C, logic header	THT / hand solder
Enclosure	Extruded aluminium case with end panels	Machining/printing of end panels, box build
Accessories	Two 10× probes + USB cable	Kitting
Test + calibration	Offset/gain per channel, supply accuracy, DMM against reference	Fixture-based FCT, operated by factory

Mixer's production design already assumes a 100-piece JLCPCB board + assembly share (\$8). That makes Mixer a standard EMS product: any competent Shenzhen turnkey PCBA house with box-build can take it end to end.

3 Target operating model: who does what

The factory owns “making it identically every time”. Protopulse owns “what it is, how it is tested, and whether it may ship”.

Activity	Protopulse India	China CM	Notes
Product architecture, R&D;, next generation	●		
Production-ready design (DFM/DFA changes)	●	●	Factory proposes, Protopulse approves by ECN
Manufacturing package, BOM, AVL	●	verify	Single source of truth, revision controlled
Component sourcing + IQC	approve A-class	●	Critical parts only from AVL
PCB fabrication, SMT/THT, AOI		●	
Sheet metal, CNC, acrylic, harness		●	Subcontracting allowed only if declared
Module / box build		●	Baker: SKD modules first
Firmware flashing	supplies locked image	●	No source code at factory
Calibration + 100% functional test	owns rig, limits, software	●	Operated by factory, data to Protopulse
FQC / OQC, AQL cosmetic	defines	●	
Packaging	designs	●	Arrives in India ready to sell
Export, freight booking	chooses forwarder	FOB	Protopulse forwarder from FOB Shenzhen
Import, customs, GST	●		Protopulse is importer of record
Baker final integration (SKD phase)	●		Until turnkey is proven
Final acceptance, activation, serial registration	●		5–15 min Mixer; longer for Baker
Warehouse, sales, shipping, support	●		
Failure analysis, supplier management	●	●	Factory does 8D on its defects

● owns · ● contributes

Why not simply “make everything in China”?

- **Baker at 100 units:** the landed-cost arithmetic in \$12 leaves almost no room for factory labour unless component prices fall at volume. Shipping a crated machine also costs more than shipping a parts kit.
- **Safety-critical calibration:** Baker carries a Class 4-power laser and a 500 W spindle. Until the factory has shown PVT yield, the last safety and make-a-PCB check stays in Delhi.
- **IP:** keeping integration, calibration logic and activation in India means no single factory holds the whole product.
- **Mixer is the opposite case:** small, light, standard EMS work, with duty and freight a small share of cost. Full turnkey is the obvious choice.

4 Before any factory sees a file: design freeze

Factories quote what you send. If the design moves after RFQ, every quote, DFM report and fixture moves with it. Close these items first (Phase 0, weeks 0–6).

#	Item	Product	Why it blocks China	Exit evidence
F1	PR-001 mask cleanliness on 0.5 mm pitch	Baker	Defines Zone B process + test board	Photo record, pass on test board
F2	PR-002 film registration $\pm 50 \mu\text{m}$ after clamping	Baker	Sets clamp frame drawing + calibration method	Fiducial measurement data
F3	PR-003 jig-to-pin accuracy after resin shrink	Baker	Decides jig process (separate printer vs UV-engine vat)	Measured pocket error, scale factor
F4	PR-004 laser offset drift with temperature	Baker	Sets calibration interval + factory calibration step	Cold/hot offset data
F5	Board size: 100 × 160 (workflow) vs 100 × 150 mm (BOM)	Baker	Magazine, fixture plate, jig tooling	Signed decision + drawings
F6	Jig printing on Baker's own UV engine (flagged for testing)	Baker	Removes or keeps a whole sub-assembly	Working vat + lift, print accuracy data
F7	Production electronics: Manta M8P + CB1 (off-the-shelf) or custom Protopulse controller	Baker	Harness, control box, test fixture all depend on it	Frozen schematic/wiring diagram
F8	Production Mixer PCB (8-bit AD9288/DAC908, FT232HL, Tang Nano 9K) built and verified	Mixer	Prototype uses modules; production is a new board	Rev A board passing full test
F9	Front-end tuning finished on 12-bit prototype, carried to production	Mixer	Analog limits in the test spec	Frozen limits table
F10	Production test plan + limits for both products	Both	Factory FCT is built from it	Test spec Rev A
F11	Regulatory classification (see §16)	Both	May force design changes (laser safety, radio)	Consultant memo
F12	Packaging concept + drop/transport protection for Baker	Both	Crate size drives freight cost	Packaging drawing, dims, weights

5 BOM control: approved vendors and substitution classes

Every BOM line gets a class. The class, not the factory's buyer, decides whether a part may be substituted.

Class	Rule	Baker examples	Mixer examples
A No substitution	Exact MPN from AVL only. Any change needs a Protopulse ECN and re-qualification.	Manta M8P V2, CB1, 10.1" 12K mono LCD + ChiTu L K1, 405 nm UV board, 10 W 445 nm laser module, spindle, Mean Well LRS-350-24, laser-safe window, interlocks, E-stop	Tang Nano 9K, AD9288, DAC908, FT232HL, ADS1256, AD8066, REF5025, INA226
B Approved alternates	Only alternates listed in the AVL; factory must notify before first use.	NEMA17/23 motors, SFU1605 screws, HGR15 rails, drag chains, SSR, thermocouple amp, pumps, solenoids, electromagnets	Relays, protection parts, USB-C, BNC, banana jacks
C Factory choice	Factory may choose, to Protopulse drawing/spec.	Fasteners, brackets to drawing, cable ties, labels, foam	Passives to spec, screws, packaging inserts

Master BOM columns the factory receives

Ref/designator · Protopulse part number · Description · MPN · Manufacturer · Qty per unit · Class (A/B/C) · Approved alternates · Lifecycle status · Critical characteristic · Drawing/spec reference · Supplier (for A-class) · Target price at 100/500/1,000.

Supply-chain risk matrix (starting view)

Part	Risk	Why	Mitigation
BTT Manta M8P + CB1	High	Single-maker hobby-market boards; revisions change without notice	Buy direct from BIGTREETECH in volume, freeze revision, hold safety stock; long-term: custom controller
10.1" 12K mono LCD	High	Consumable-grade LCD, model churn in resin-printer market	Qualify 2 sources; design mount for 2 panel variants; spares policy
Laser module	High	Safety-critical, power/optics vary by batch	Incoming power check 100%; single approved maker
Tang Nano 9K / GW1NR-9	Medium	Module availability	Move to chip-down FPGA on the main PCB at volume
AD9288, DAC908, ADS1256, AD8066	Medium	Analog parts, counterfeit risk in grey market	Authorised channel only (e.g. LCSC/authorised distributor), lot traceability
Spindle	Medium	Retail kit today; runout varies	Factory sources industrial version, runout spec at IQC
Rails, screws, motors	Low-Med	Many makers, quality spread	IQC straightness/backlash, 2 approved makers
Sheet metal, acrylic	Low	Local to factory	Drawings with tolerances, first-article inspection

6 The manufacturing master package

This folder is the single source of truth. Every file carries a revision; nothing is sent as “final_v7”. One package per product (BAKER-MP-RevA, MIXER-MP-RevA).

```

PROTOPUS_BAKER_MP_RevA/
├── 00_MASTER      product overview · revision history · golden-sample record · this plan
├── 01_ELECTRONICS control-box wiring diagram · harness drawings · (custom controller: schematic,
                  Gerber/ODB++, CPL, assembly drawing, stack-up)
├── 02_BOM         master BOM (A/B/C classes) · AVL · alternates · critical parts list
├── 03_MECHANICAL  STEP + 2D drawings with tolerances · sheet-metal flat patterns · finishes
                  · shuttle fixture plate · registration pins · laser-window spec
├── 04_FIRMWARE   locked factory image (binary only) · flashing procedure · version table
├── 05_ASSEMBLY   work instructions per station · torque table · cable routing · adhesives
├── 06_TEST        FCT spec + limits · Mixer test-rig manual · calibration procedures
                  · make-a-PCB acceptance job (test Gerber) · safety tests
├── 07_QUALITY    IQC / IPQC / FQC / OQC plans · AQL table · defect catalogue with photos
├── 08_PACKAGING  crate / carton drawings · foam · labels · manuals · accessory kit list
├── 09_COMPLIANCE classification memo · test reports · marking artwork

PROTOPUS_MIXER_MP_RevA/
├── 01_ELECTRONICS schematic (PDF) · Gerber/ODB++ · drill · CPL · assembly drawing · stack-up
├── 02_BOM         BOM with MPNs, classes, alternates
├── 03_MECHANICAL  extrusion + end-panel drawings · front-panel artwork
├── 04_FIRMWARE   FPGA bitstream + FT232H config (binary) · programming procedure
├── 06_TEST        bed-of-nails fixture drawing · FCT limits · calibration procedure
├── ... same 00, 05, 07, 08, 09 structure

```

What the factory must NOT receive

- Firmware and FPGA source code, Proto source, calibration algorithms, test-software source.
- Customer lists, pricing, roadmap, or the design rationale for proprietary process steps beyond what assembly needs.
- Full native CAD for parts they do not make (send STEP/2D for what they make; black-box the rest).

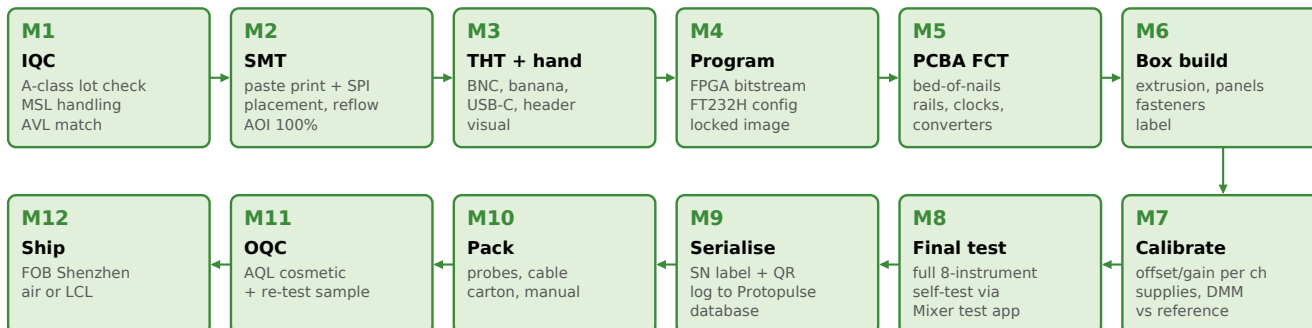
Golden samples

Build and fully test golden units in Delhi, label them “GOLDEN SAMPLE, NOT FOR SALE”, sign and photograph them. Suggested: **Mixer** 3 units (factory, Protopulse, spare/inspector); **Baker** 1 unit at the factory once the SKD/turnkey supplier is chosen, plus one reference set of modules. Every production unit is compared against the golden unit and the golden test log.

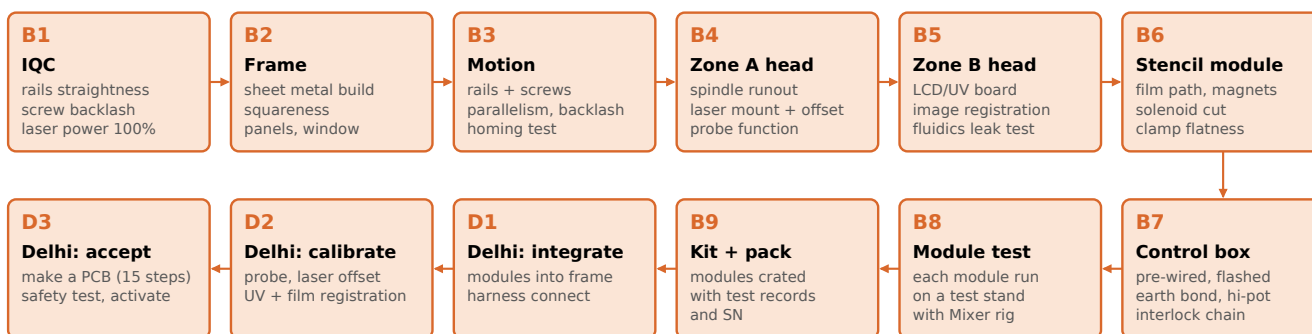
7 The assembly lines: station by station

These are the lines Protopulse asks the factory to set up. Station content is fixed by Protopulse; takt times and staffing come from the factory's PVT data.

Mixer line (full turnkey, EMS)



Baker line, stage 1: tested modules (SKD) in China, integration in Delhi



Baker line, stage 2 (full turnkey): the same stations B1-B8 plus D1-D3 move into the factory as B9 Integrate → B10 Calibrate → B11 Make-a-PCB acceptance job → B12 Burn-in (motion cycles, thermal) → B13 Safety test (earth bond, insulation, interlocks, E-stop, laser cut-off on door open) → B14 Crate → B15 OQC. Delhi keeps a shortened acceptance: power-on, homing, calibration check against a fiducial board, activation.

8 Potential factories in China and how to choose

Candidates below are companies whose public websites describe the relevant capability (as found in the 29 Sep 2026 research). They are a starting longlist, **not endorsements**. Capability, ownership of equipment, and fit must be proven by questionnaire, video audit, third-party audit and a paid pilot build.

A. EMS / turnkey PCBA + box build (Mixer; Baker electronics)

Company (site)	Publicly stated capability	Best fit for	Stage
Sseed Studio · Fusion (sseedstudio.com)	PCB, sourcing, SMT/THT, functional test, DFM/DFA, kitting, packaging; no-MOQ PCBA; published flow IQC → SMT → FCT → assembly → FQC	Mixer turnkey, 10 → 1,000 units	Start
PCBWay (pcbway.com/oem)	PCBA + CNC, sheet metal, injection, integrated assembly, box build, testing	Mixer; Baker electronics + sheet metal	Start
JLCPCB (jlcpcb.com)	PCB + PCBA at low volume (already in the Mixer cost model)	Mixer PCBA for EVT/DVT; cost benchmark	Start
Peakingtech (peakingtech.com)	Startup-focused: NPI from ~5 units, production from ~50; SMT/THT, box build, AOI, FCT, enclosures, molding	Mixer turnkey; Baker SKD candidate	Start
Shenzhen Shenze Electronic (shenzeelectronic.com)	PCB, sourcing, SMT/THT, AOI, X-ray, ICT, FCT, coating, box build; prototype to medium volume	Mixer turnkey	Start
PCBONLINE (pcbonline.com)	PCB, SMT/THT, sourcing, IC programming, thermal ageing, box build, finished-product test, shipping	Mixer turnkey	Start
Grandtop (grandtopgroup.com)	Prototype from 1 pc, SMT, box build, system integration; industrial/medical focus	Mixer; industrial-grade builds	Start
Aigelin (aigelin.org.cn)	SMT → wave → box build → test; 3 China plants; 0→1 to volume	Mixer at volume; Baker electronics	Scale

B. Mechatronics / integrated mechanical + electronics (Baker)

Company (site)	Publicly stated capability	Best fit for	Stage
Shenzhen MHC (mhcodm.com)	CNC, moulding, SMT/THT, final assembly, box build, NPI, design validation, fixture preparation	Baker SKD → turnkey (lead candidate)	Start
Shenzhen Zhendatech (zhendatech.com)	SMT/DIP, coating, harness, sheet metal, injection, complete-product manufacturing; ISO 9001/13485, IATF 16949	Baker turnkey once mature	Start
Shenzhen ShockTech (szshock.com)	Moulding, injection, painting, hardware machining, SMT, hand solder, assembly, test, packaging	Baker frame + covers + assembly	Start
Inno Manufacturing (innomanufacturing.com)	Plastic + metal fabrication, electromechanical products, full assembly, test, packaging, logistics; startup-focused	Baker SKD / turnkey	Start
Mersain (mersain.com)	Mechanical + electrical engineering, firmware, molding, PCBA, EVT/DVT/PVT, mass production	Baker if design help is wanted (watch IP scope)	Option
PCBWay (see above)	CNC + sheet metal + assembly on one platform	Baker frame/brackets for EVT	Start

C. Large EMS for later scale (10,000+ units/yr)

Company (site)	Publicly stated capability	When
Shenzhen Kaifa Technology (en.kaifa.cn)	Major EMS: procurement, PCBA, finished-product assembly, system integration, testing, logistics, traceability; DFMEA/PFMEA/DFM/DFT; Shenzhen + Dongguan sites	When Mixer volume justifies a tier-1 EMS
3CEMS (3cems.com)	JDM, prototype, NPI, MP, PCBA, box build, system integration, test, logistics, component traceability	Scale-up / second source

D. Direct component makers to buy from in volume

For A-class parts, ask the CM to buy from (or let Protopulse negotiate directly with) the maker: BIGTREETECH / BIQU (Manta M8P, CB1, HDMI7), ChiTu Systems (10.1" LCD + L K1 board), StepperOnline / OMC (NEMA17/23, Mean Well at 100+), LCSC and authorised distributors (ICs), and the laser and spindle makers the CM proposes during DFM. Volume quotes on the spindle, laser and LCD are where the Baker turnkey business case is won or lost (\$12).

Selection funnel

Step	Mixer track	Baker track	Output
Longlist	6 EMS from table A	6 from table B (+ PCBWay)	Same RFQ package to all, independent quotes
Questionnaire + NDA/NNN	All	All	Completed questionnaire (Appendix B)
Video audit (live, unscripted)	Top 4	Top 4	Warehouse → SMT → AOI → assembly → test → QC shown live
DFM review	Top 3	Top 3	Written DFM report + open-book quote
On-site / third-party audit	Top 2	Top 2	Audit by SGS, TÜV, Bureau Veritas or Intertek, or own visit
Paid EVT pilot	2 suppliers × 10 units	2 suppliers × module sets	Compare yield, substitutions, documentation, communication
Award	1 primary + 1 backup	1 primary + 1 backup	Manufacturing + quality agreements signed

Supplier scorecard (weights differ by product)

Criterion	What “good” looks like	Mixer weight	Baker weight
Engineering / DFM	Specific, numbered DFM findings; proposes test points and fixture	20%	20%
Quality system	ISO 9001, SPI/AOI/X-ray, FCT, MES, lot traceability, calibration records	20%	15%
Mechanical capability in-house	Sheet metal, CNC, assembly benches, alignment skills	5%	20%
Landed cost (not unit price)	Open-book quote; tooling/NRE clear; volume curve	20%	15%
Supply chain	Authorised sourcing, obsolescence handling, alternates discipline	15%	10%
Communication / PM	English engineer contact, response < 24 h, weekly report	10%	10%
Stability + IP posture	Legal entity, years operating, accepts NNN + no-subcontract clause	10%	10%

9 NPI gates: EVT → DVT → PVT → MP

Quantities are sized to each product's value. Baker units are expensive, so Baker gates use fewer machines and more module-level testing; Mixer gates follow the classic pattern.

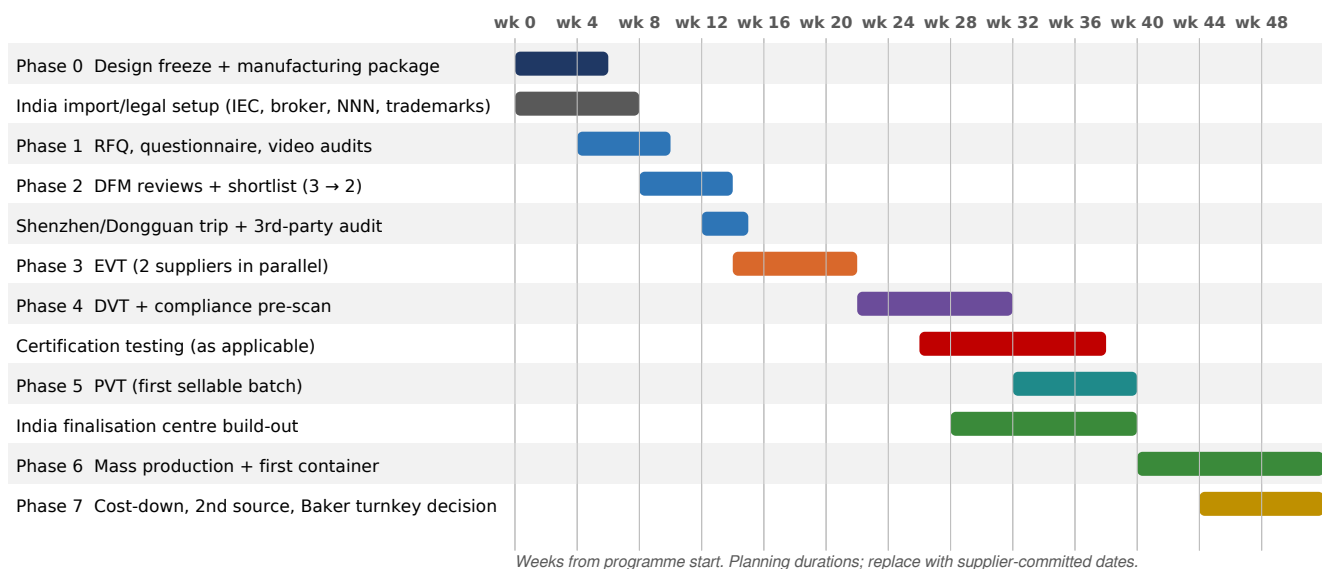
Gate	Mixer qty	Baker qty	Question it answers	Exit criteria (all must pass)
EVT	10 per supplier (2 suppliers)	2-3 module sets per supplier; 1-2 integrated in Delhi	Can the factory reproduce the prototype?	All units meet test spec; DFM actions closed or accepted; no unapproved substitutions; BOM and drawings updated to Rev B
DVT	50	5 machines	Is the design right for production and certification?	Reliability + thermal + transport (drop/vibration) tests pass; EMC/safety pre-scan clean; Baker open points PR-001...004 pass on factory units
PVT	100 (first sellable batch)	10-20 machines	Is the production process right?	Built by line operators with line fixtures; first-pass yield ≥ target (set at DVT, e.g. ≥ 95%); packaging passes; test data uploading; signed Production Authorisation
MP	Rolling orders	Rolling orders	Can we scale safely?	Monthly yield + field-return review; ECN discipline; second source qualified

Production authorisation (signed at the end of PVT)

```

PRODUCTION AUTHORISATION                                PA-MIXER-001
Product ..... Mixer                                    Revision ..... Rev C
Factory ..... [primary CM]                             Backup ..... [backup CM]
PVT quantity ... 100                                   First-pass yield ..... %
Functional test 100% (Mixer test app v_) Calibration ..... 100%
Golden sample .. approved __/__/2027                   Packaging ..... approved
Firmware ..... bitstream __ / FT232H cfg __ (locked image hash __)
Open ECNs ..... none / listed
Production ..... AUTHORISED   Signed: Protopulse Engineering + Quality
  
```

Timeline (about 12 months to first container)



Mixer can run ahead of Baker: its EVT can start as soon as the production board (F8) passes, while Baker is still closing its open points.

10 The production test system (Protopulse owns it)

The factory runs the tests; Protopulse owns the rig design, the software, the limits and the data. If the factory changes, the test system moves with Protopulse.

Architecture

- **Mixer as the tester.** A production Mixer plus a Protopulse test app drives supplies, stimulus, scope/logic capture and DMM readings on the unit under test. For Mixer production, a golden Mixer and a bed-of-nails fixture test each new Mixer; for Baker, the Mixer rig tests the control box and each module on a test stand.
- **Test app:** scan serial → run sequence → PASS/FAIL per step with measured value and limit → upload to the Protopulse database. Operators cannot edit limits.
- **Database:** one record per serial: PCB lot, component lots (A-class), firmware/bitstream hash, calibration constants, every test value, operator, station, timestamp, rework history.
- **Golden data:** golden-sample logs define the reference; drift in the population triggers investigation before it becomes failures.

Serial numbers and traceability

Format: **PP-BKR-YYYY-NNNNNN** and **PP-MXR-YYYY-NNNNNN** (e.g. PP-MXR-2027-000173). Baker modules carry sub-serials (PP-BKR-ZA-..., -ZB-..., -CB-...) linked to the machine serial at integration.

What “100% tested” means (example rows from the test spec)

Product	Test	Method	Limit source	Sample
Mixer	Supply outputs accuracy + current readback	Fixture DMM vs INA226 reading	Test spec Rev A	100%
Mixer	Scope channel offset/gain	Known reference levels from fixture	Calibrate then verify	100%
Mixer	Wavegen amplitude/frequency	Captured by fixture / golden Mixer	Test spec	100%
Mixer	16-ch logic + pattern loopback	Loopback cable, all channels	Bit-exact	100%
Mixer	DMM accuracy	Against fixture reference (REF5025-class)	Test spec	100%
Baker	Earth continuity + insulation (hi-pot)	Safety tester	Applicable safety standard	100%
Baker	Door interlock + E-stop cut laser, spindle, motion	Open door / press E-stop during motion	Must stop, zero tolerance	100%
Baker	Homing repeatability, backlash per axis	Dial indicator / probe	Test spec	100%
Baker	Spindle runout, laser power, laser-to-spindle offset	Indicator, power meter, fiducial cut	Test spec	100%
Baker	UV image registration, film registration	Fiducial board	±50 µm film (PR-002)	100%
Baker	Make-a-PCB job (all 15 steps) on test Gerber	Inspect board, then test it on Mixer	Defect catalogue	100% (DVT/PVT) → sample in MP

11 Quality plan

Four QC gates

Gate	Where	Mixer checks	Baker checks
IQC	Factory receiving	A-class lot + traceability, MSL, PCB fab report, counterfeit screen on analog ICs	Rail straightness, screw backlash, motor spec, laser power 100%, LCD dead pixels, sheet-metal first article
IPQC	On the line	SPI, AOI 100%, X-ray where needed, first-article per shift, programming log	Torque audit, rail parallelism, harness continuity, cable routing, module tests
FQC	End of line	100% calibration + final test, cosmetic	100% safety + functional + calibration, make-a-PCB (per plan)
OQC	Before shipment	AQL cosmetic/packaging + re-test sample; photos	Crate check, accessories, documents, photos; pre-shipment inspection (third party at first batches)

Defect classes

Class	Definition	Examples	Acceptance
Critical	Could injure a person or damage property	Laser firing with door open, missing earth bond, E-stop not stopping motion, exposed mains	Zero. Stop the line, 8D, containment of all units
Major	Unit does not meet function or spec	Channel out of spec, axis not homing, UV image offset, wrong firmware	Zero at FQC (100% tested); AQL for OQC re-test
Minor	Cosmetic or documentation	Scratch, label skew, manual missing page	AQL per quality agreement

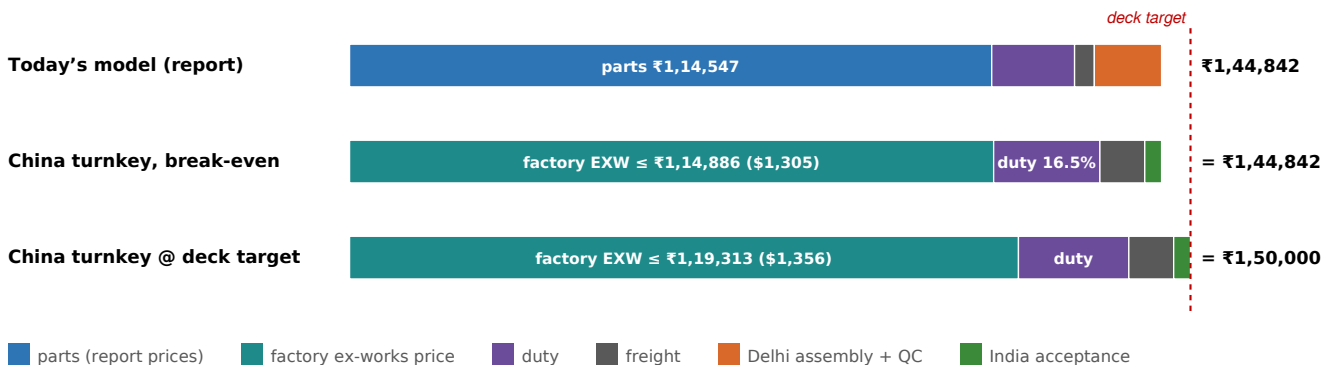
Quality agreement: minimum contents

- Scope, product revisions, and the manufacturing package as the controlling document.
- Incoming, in-process and final inspection plans (this section), sampling tables, record retention (≥ 3 years).
- Change control: no change to part, supplier, material, process, site, firmware or packaging without Protopus written approval (ECN).
- Non-conformance handling: scrap, rework rules, 8D reports within agreed days, containment, and cost of failure.
- Rejected units: serial-level disposition, destruction with photos for proprietary parts, no resale of rejects.
- Right to audit (announced and unannounced), right to third-party inspection before shipment.
- Warranty: factory liability for workmanship defects and for components it sourced.

12 Landed cost: when does China turnkey win?

Today's production figures are **China parts + Delhi assembly**. Moving assembly to China does not delete the ₹12,000 Delhi assembly and the ₹12,000 Indian sheet-metal frame; it replaces them with factory labour, overhead and margin, and adds duty on that value and a bigger freight bill for a crated machine. Compare landed Delhi cost only.

Baker



Break-even assumes ₹8,000/unit freight for a crated machine and ₹3,000/unit India acceptance (assumptions, replace with quotes). Duty rate from the cost report; HS code to confirm.

Line	Today (report)	China turnkey (formula)
Parts	₹1,14,547 (\$1,302)	inside factory EXW price
Duty	₹14,775 (16.5% on imported parts)	16.5% × EXW (rate to confirm for a complete machine's HS code)
Freight	₹3,520 (LCL parts share)	≈ ₹8,000 crated machine (assumption)
India labour	₹12,000 assembly + QC	≈ ₹3,000 acceptance + activation (assumption)
Landed	₹1,44,842	EXW × 1.165 + 8,000 + 3,000

What this means

To match ₹1,44,842, a finished Baker must cost ≤ **₹1,14,886 (\$1,305) ex-works**; to stay at the ₹1,50,000 target, ≤ **₹1,19,313 (\$1,356)**. Parts alone cost about **\$1,302** at the report's prices. So at 100 units, turnkey only pays if the factory buys parts well below the report's prices, mainly the retail-priced spindle (\$85.99), laser (\$159) and LCD (\$110.99), and China sheet metal. That is why Baker starts as SKD and moves to turnkey only when quotes prove it.

Mixer

Line	Today (report)	China turnkey (formula)
Parts (incl. \$8 PCB + assembly share)	₹9,926 (\$112.8)	inside factory EXW price
Duty	₹1,638	16.5% × EXW (confirm HS code for a finished instrument)
Freight	₹176	≈ ₹176 (small carton; confirm)
India labour	₹2,000 assembly + QC	≈ ₹500 acceptance + activation (assumption)
Landed	₹13,740	EXW × 1.165 + 676
Break-even EXW		≤ ₹11,214 (\$127) to match today · ≤ ₹12,295 (\$140) at ₹15,000 target

Mixer leaves the factory about \$15–27 over parts for box build, test, packing and margin: tight at 100 units, comfortable at 1,000. Get the volume curve in the RFQ.

Ask every factory for four quotes

Quote	China scope	India scope	Use
A CKD	Parts kits only	All assembly	Benchmark vs today's model
B SKD	Tested modules	Integration + acceptance	Baker stage 1
C Complete, untested	Full assembly	Test + calibration	Rarely right; exposes test cost
D Turnkey	Sourcing → test → pack	Acceptance + activation	Mixer from day 1; Baker stage 2

For each quote, request prices at 5 / 10 / 25 / 50 / 100 / 250 / 500 / 1,000 / 5,000 units, split open-book: PCB, PCBA, components, mechanical parts, tooling, NRE, labour, test, calibration, packaging, overhead/margin, plus carton/crate size, gross weight, MOQ, lead time, payment terms, EXW and FOB.

13 Change control, tooling and fixtures

Engineering change notice (ECN)

ECN-004 Product: BAKER Status: APPROVED
 Old: SHUTTLE_PLATE Rev B New: SHUTTLE_PLATE Rev C
 Reason: registration-pin position change after PR-003 jig shrink data
 Effective: DVT build onward (serials PP-BKR-2027-000006 →)
 Stock disposition: rework 12 plates, scrap 0
 Impact: jig model scale, calibration procedure CAL-03 Rev B
 Approved: Protopulse Engineering ____ Quality ____ Factory ack ____

- The factory may **request** changes (ECR); only Protopulse **approves** them (ECN).
- Component end-of-life: factory notifies → Protopulse evaluates alternate → prototype + test → ECN → BOM revision → production. Never “equivalent part used”.
- Every ECN names the effective serial number so field failures can be traced to the right revision.

Tooling and fixtures

Item	Owner	Rule
Sheet-metal tooling, bending dies, extrusion dies, molds (if any later)	Protopulse (paid)	Tool ID, drawings, cavity count, expected life, maintenance log, storage at factory, transfer on 30 days' notice
Mixer bed-of-nails fixture, Baker module test stands, calibration fixtures	Protopulse (designed in India, built by factory or India)	Duplicate set kept in Delhi; software licensed per station
Golden samples, reference boards, fiducial boards	Protopulse	Serialized, signed, re-verified each batch
Test app + limits	Protopulse	Installed as locked build; updates only from Protopulse

Use bridge tooling first: laser-cut and bent sheet metal and machined parts through EVT/DVT; commit to hard tooling only after DVT freezes the geometry.

14 Protecting the IP

Contracts (have a China-qualified lawyer draft them)

Document	Purpose	Key clauses
NNN agreement	Non-disclosure, non-use, non-circumvention	Written in Chinese (with English), governed by PRC law, enforceable in a Chinese court near the factory, liquidated damages. Sign before the full package is shared
Manufacturing agreement	Commercial relationship	Pricing + volume curve, forecast vs firm orders, lead times, payment milestones, IP and tooling ownership, no subcontracting without consent, no overruns, no sale of Protopulse products or rejects to anyone else, termination + tooling transfer
Quality agreement	How quality is achieved and proven	See §11
Tooling agreement	Who owns what	Asset list with IDs and photos; return on demand

Architecture (the protection that works even if a contract fails)

- **Register trademarks in China early** (China is first-to-file): ProtoPulse, Protopus, Baker, Mixer, Proto, in the relevant classes. Consider design patents for distinctive hardware.
- **Locked factory image.** The factory flashes binaries only; no source for firmware, FPGA or Proto. Where the hardware allows, enable read-out protection / secure boot.
- **Activation in India.** Proto features unlock at India acceptance against the serial in the Protopulse database. A unit that skips India cannot be sold as a working product.
- **Calibration logic stays with Protopulse.** The test app computes and stores calibration constants; the algorithm is not documented to the factory.
- **Split where it matters.** In Baker SKD, no single supplier builds all modules and integrates them. Keep the most process-specific item (stencil/film module or Zone B head) with a second supplier or in India if needed.
- **Count units.** Serial numbers issued by Protopulse only; the factory reports scrap by serial with photos.

15 China → Delhi logistics and India import setup

Set up once, before the first commercial shipment

Item	What it is	Who
IEC (Importer-Exporter Code)	Required by DGFT for import/export activity	Protopulse via DGFT portal
GST registration + IGST credit set-up	Import IGST is creditable; the cost model assumes it is credited	CA
Customs broker (CHA)	Files the bill of entry, advises HS codes, duty and exemptions	Appoint 1 in Delhi (ICD/air cargo)
AD code registration	Bank authorised-dealer code registered at the port/ICD of import	Bank + broker
HS classification memo	Baker (complete machine vs parts) and Mixer (measuring instrument) may fall under different headings and rates than the HS 8501 basis used in the report	Broker + CA
Freight forwarder	Protopulse chooses; compare 2-3 quotes; FOB Shenzhen terms	Protopulse

Routing

Shipment	Mode	Typical route	Notes
Samples, EVT/DVT, urgent spares	Air courier / air freight	Shenzhen/Hong Kong → IGI Delhi	Fast; high cost per kg; fine for Mixer
Mixer batches	Air freight or LCL	Yantian/Shekou → Nhava Sheva or Mundra → ICD Delhi (rail), or air	Small and light; compare per-unit cost at each batch size
Baker crates / modules	LCL → FCL as volume grows	Yantian/Shekou → Nhava Sheva or Mundra → ICD Tughlakabad/Patparganj	Report uses LCL all-in (Sino-Shipping, Sep 2026); confirm transit time with forwarder

Incoterms and payment

- Ask for EXW, FOB and CIF in every quote; buy **FOB** with your own forwarder so the manufacturing price and the logistics price stay separate and auditable.
- Be careful with DDP offers: they can hide who is importer of record, the declared HS code and value. Protopulse should be the importer of record with clean documents.
- NPI/tooling: pay by milestone (tool made → first samples → approval). Production: a deposit, with the balance after pre-shipment inspection and before release, is common; negotiate the split. Avoid 100% upfront.

16 Regulatory and safety checklist

Classify before PVT, not after 1,000 units exist. Items below are questions to put to a certification consultant and the customs broker; they are not conclusions.

Topic	Why it applies	Product	Action
Laser safety	10 W (9.5 W optical) 445 nm diode laser is Class 4 power; the enclosed machine with interlocks and a laser-safe window aims to be safer overall (IEC 60825-1 classification of the complete product)	Baker	Classify the product; interlock design review; labels; test interlocks 100%; goggles in the box for service
Machine electrical safety	Mains-powered machine with spindle, heater (300 W), motors	Baker	Design to an applicable machine-safety standard (e.g. IEC 60204-1 for electrical equipment of machines); earth bond + hi-pot at FQC
Wireless	The BTT CB1 compute module includes Wi-Fi	Baker	Check WPC Equipment Type Approval (ETA) requirements with the module as used
BIS compulsory registration	Some electronics/IT products are under BIS CRS	Both	Check whether either product (or its power supply) falls in a notified category
E-waste EPR	E-Waste Management Rules apply to listed electrical/electronic equipment categories	Both	Check category; register as producer/importer if required
Legal Metrology	Pre-packaged goods declarations for importers	Both	Importer label on the box
EMC	Switching supplies, motors, FPGA	Both	Pre-scan at DVT; full test if targeting export markets
Export markets (later)	CE (EMC, low voltage, machinery), FCC, RoHS	Both	Design for it now; certify when entering those markets

17 The India finalisation centre

India is not a second factory. It is a small, clean finalisation and distribution centre.

Area	Equipment	Purpose
Receiving + quarantine	Pallet space, barcode scanner, laptop	Scan every serial; untested units never mix with accepted ones
Mixer acceptance bench	ESD mat, golden Mixer, test fixture, laptop with test app	Power-on, quick self-test, activation, database registration (target 5-10 min)
Baker integration + acceptance bay (SKD phase)	Assembly bench, lifting aid, torque tools, dial indicator, laser power meter, fiducial boards, safety tester, Mixer rig, fume/dust extraction	Integrate modules, calibrate, run the make-a-PCB job, safety test, activate
Failure analysis bench	Scope, DMM, bench supply, rework station, microscope	Diagnose returns, feed 8D to the factory
Stores + dispatch	Racking, packing table	Accepted stock, accessories, customer shipping

India acceptance flow



People

- Year 1: 1 manufacturing/quality engineer (owns suppliers, ECNs, test system), 1-2 technicians (acceptance, Baker integration in SKD phase), part-time logistics/stores.
- Visit the factory for EVT, DVT and PVT builds; weekly video review with the factory PM; monthly quality review in MP.

18 Programme budget: what is known and what to quote

Unit-based lines use today's production unit costs as a **floor**; NPI builds usually cost more per unit than mass production. Everything else must come from quotes.

Item	Basis	Floor estimate	Get from
Mixer EVT: 2 suppliers × 10 units	20 × ₹13,740	₹2,74,800	Supplier quotes
Mixer DVT: 50 units	50 × ₹13,740	₹6,87,000	Supplier quotes
Mixer PVT: 100 units (sellable)	100 × ₹13,740	₹13,74,000	Supplier quote; recovered by sales
Baker EVT: module sets, 2 suppliers	≈ 2-3 machine-equivalents × ₹1,44,842	₹2,89,684 – ₹4,34,526	Supplier quotes
Baker DVT: 5 machines	5 × ₹1,44,842	₹7,24,210	Supplier quotes
Baker PVT: 10-20 machines (sellable)	10-20 × ₹1,44,842	₹14,48,420 – ₹28,96,840	Supplier quote; recovered by sales
Golden samples	3 Mixer + 1 Baker at prototype cost	₹2,24,541 + ₹3,73,418	Already costed (prototype)
Test fixtures + rigs	Bed-of-nails, module stands, calibration fixtures	To quote	Fixture house / factory
NRE + tooling	Sheet metal, extrusion end panels, jigs	To quote	Open-book RFQ
Audits + inspection	Third-party factory audit ×2-4, pre-shipment inspections	To quote	SGS / TÜV / BV / Intertek
Travel	Delhi ↔ Shenzhen trips for EVT/DVT/PVT	To quote	Travel desk
Legal + IP	NNN, manufacturing/quality agreements, China trademarks	To quote	China-qualified law firm
Compliance testing	Per §16 classification	To quote	Test lab / consultant
Freight for NPI units	Air for EVT/DVT	To quote	Forwarder

19 Risk register

#	Risk	L	I	Mitigation	Owner
1	Baker turnkey quotes don't beat Delhi build at low volume	H	M	SKD first; four-quote RFQ; revisit at 250-500 units/yr	Ops
2	Open Baker process points fail on factory units	M	H	Close PR-001...004 in Phase 0; re-verify at DVT	Eng
3	Hobby-market boards (Manta, CB1, LCD) change revision or go EOL	M	H	Freeze revision, safety stock, second LCD source, custom controller roadmap	Eng
4	Unapproved substitution by factory	M	H	A/B/C classes, AVL, lot traceability, IQC audits, contract penalties	Quality
5	Laser safety incident or non-compliant classification	L	Critical	100% interlock test, classification by consultant before PVT	Eng
6	IP leak / copycat product	M	H	NNN, China trademarks, locked images, activation in India, split modules	CEO
7	Duty or HS code differs from the cost model	M	M	Broker classification memo before PVT; binding ruling if needed	Finance
8	Headroom to deck targets is thin (Baker 3.4%, Mixer 8.4%)	H	M	Volume quotes on spindle/laser/LCD; value engineering at DVT	Eng + Ops
9	Single-factory dependence	M	H	Backup supplier qualified by PVT; fixtures and tools owned by Protopulse	Ops
10	Transport damage to Baker	M	M	Crate design + transport test at DVT; insurance	Ops
11	Test escapes (units pass factory test, fail in field)	M	H	India acceptance, golden-data trend monitoring, field-return 8D	Quality
12	Currency movement (USD/INR, CNY)	M	M	Quote in USD with FX clause; review at each PO	Finance

L = likelihood, I = impact (L/M/H).

20 Action plan: the next 90 days

When	Action	Output
Week 1-2	Close board-size decision (F5); start PR-001...004 test runs; freeze Mixer production board layout	Decisions signed; test plans
Week 1-2	Apply for IEC; appoint CA + customs broker; request HS classification memo	IEC; broker engaged
Week 2-3	File China trademark applications; engage China-qualified lawyer for NNN + agreements	Filed marks; NNN draft
Week 2-4	Build manufacturing packages Rev A (§6), BOM classes, test spec Rev A	BAKER-MP-RevA, MIXER-MP-RevA
Week 4	Send RFQ (Appendix A) + questionnaire (Appendix B) to 6 EMS + 6 mechatronics suppliers	12 RFQs out
Week 5-7	Live video audits of top 4 per track; collect DFM reports + open-book quotes	Scored shortlist
Week 6	Build golden samples (3 Mixer, 1 Baker) + test rigs v1	Signed golden units
Week 8-10	Shenzhen/Dongguan trip: 4-6 factories; third-party audit on finalists	Top 2 per track
Week 10-12	Place paid EVT orders with 2 suppliers per track; sign NNN + draft quality agreement	EVT underway

Suggested Shenzhen / Dongguan visit plan

Day	Plan
1	Shenzhen: 2 EMS candidates (Mixer). Walk the line from IQC to OQC; watch AOI and FCT live.
2	Shenzhen: 2 more EMS or component makers (BIGTREETECH, ChiTu) for volume pricing on A-class parts.
3	Dongguan: 2 mechatronics / sheet-metal suppliers (Baker). See alignment benches, sheet-metal shop, assembly.
4	Finalist deep-dive: DFM workshop with golden samples; agree EVT scope; third-party auditor on site if booked.

Appendix A RFQ email template

Subject: RFQ – Protopulse [Mixer / Baker] – turnkey manufacturing, 5 to 5,000 units

Dear [Name],

ProtoPulse (Delhi, India) designs electronics prototyping equipment. We are selecting a manufacturing partner for [Mixer, a USB electronics instrument / Baker, a desktop PCB-making machine]. The design is validated and a production revision is ready for DFM.

After signing our NNN agreement you will receive the manufacturing package (BOM, drawings, Gerber/CPL where applicable, assembly and test specifications).

Please quote:

1. Four scopes: A) parts kits, B) tested sub-assemblies, C) complete untested, D) turnkey (sourcing, manufacturing, programming, calibration, 100% functional test, packing).
2. Quantities: 5 / 10 / 25 / 50 / 100 / 250 / 500 / 1,000 / 5,000.
3. Open-book split: PCB, PCBA, components, mechanical parts, tooling, NRE, labour, test, calibration, packaging, overhead/margin.
4. Incoterms EXW, FOB Shenzhen, CIF Nhava Sheva; carton/crate size and gross weight.
5. MOQ, lead time (samples and production), payment terms, warranty and rework policy.
6. Which processes you perform in-house and which you subcontract (with the subcontractor).
7. A DFM/DFA review with specific findings, and a proposed EVT plan.

Please also complete the attached supplier questionnaire.

We use our own production test system and fixtures, operated by your team.

Best regards,

[Name] · ProtoPulse, Delhi · [email] · [phone]

Appendix B Supplier questionnaire

Company

1. Legal company name and registration number
2. Factory address(es); owned or leased
3. Employees (total / engineering / QA)
4. Factory area
5. Years operating
6. Main industries and three reference customers
7. Annual revenue range

Production

8. Number of SMT lines; placement machines; smallest package handled
9. THT and wave/selective solder
10. In-house CNC, sheet metal, bending, powder coat
11. Cable/harness making
12. Box-build / system integration benches
13. Functional test and calibration capability
14. Packaging design and crate making

Quality

15. ISO 9001 (certificate number); other certifications
16. IPC standards used (e.g. IPC-A-610 class)
17. SPI, AOI, X-ray, ICT, FCT: which in-house
18. MES and lot/serial traceability
19. ESD and MSL control
20. Calibration records for test equipment
21. Typical first-pass yield on new products

Supply chain

22. Turnkey sourcing: authorised distributors used
23. Counterfeit prevention process
24. Obsolescence / EOL notification process
25. Willingness to follow a customer AVL with no substitution

NPI

26. DFM/DFA report example
27. EVT/DVT/PVT experience; example timeline
28. Engineering contact with English; response time

Commercial

29. MOQ; NRE policy; tooling ownership terms
30. Payment terms for NPI and production
31. Lead times
32. Warranty and rework policy
33. Acceptance of NNN, no-subcontracting and no-overrun clauses

Appendix C Glossary

Term	Meaning
AOI / SPI / ICT / FCT	Automated optical inspection / solder-paste inspection / in-circuit test / functional test
AQL	Acceptable quality limit: statistical sampling plan, used for cosmetic/package checks, never instead of 100% functional test
AVL	Approved vendor list
BCD / SWS / IGST	Basic customs duty / social welfare surcharge / integrated GST (creditable)
CKD / SKD	Completely knocked down (parts kits) / semi-knocked down (tested modules)
CM / EMS / ODM	Contract manufacturer / electronics manufacturing services / original design manufacturer
DFM / DFA / DFT	Design for manufacturing / assembly / test
ECR / ECN	Engineering change request / notice
EVT / DVT / PVT / MP	Engineering / design / production validation test; mass production
EXW / FOB / CIF / DDP	Incoterms: ex-works / free on board / cost, insurance, freight / delivered duty paid
IEC	Importer-Exporter Code (DGFT)
IQC / IPQC / FQC / OQC	Incoming / in-process / final / outgoing quality control
LCL / FCL / ICD	Less-than-container load / full container load / inland container depot
NNN	Non-disclosure, non-use, non-circumvention agreement
NRE	Non-recurring engineering charges
8D	Eight-discipline problem-solving report

Sources: Protopus Cost Report, Protopus Baker Workflow Story and Protopus BOM Baker + Mixer workbook (ProtoPulse, 29 Sep 2026); China assembly-line research conversation (29 Sep 2026) citing supplier websites (Seed Studio, PCBWay, Peakingtech, Shenzhen MHC, Zhendatech, ShockTech, Grandtop, PCBONLINE, Shenze, Aigelin, Kaifa, 3CEMS, Inno Manufacturing, Mersain) and Indian regulator pages (DGFT IEC, DoT ETA, BIS, Legal Metrology). Duty, freight and compliance items to be confirmed with a customs broker and a certification consultant.